

UNIT 3 · MATHEMATICS FOR AI

Vectors: Concepts and Applications

Chapter 3.2 — Class XI

Instructor: Saptarshi Jana



Learning Objectives

1

Understand vectors and how they differ from scalars

2

Recognise the relevance of vectors in AI/ML

3

Perform vector addition, subtraction & scalar multiplication

4

Compute the magnitude and direction of a vector

5

Apply the dot product and interpret its results

6

Apply the cross product and its uses in AI

7

Represent vectors in 2D and 3D coordinate systems

8

Connect vector algebra to real-world applications

Scalars vs. Vectors

SCALARS

Magnitude only

- Temperature, mass, speed, time

*“The car is moving
at 60 km/h.”*

VECTORS

Magnitude AND direction

- Velocity, force, displacement, acceleration

*“The car is moving at
60 km/h towards the
north.”*

Vector Algebra in the AI Domain

1

Data Representation

Images as pixel-intensity vectors; words embedded via Word2Vec / GloVe.

2

ML Model Foundations

Linear regression & SVMs rely on dot products and norms.

3

Neural Network Ops

Forward & backward propagation are vectorised computations.

4

Distance & Similarity

Euclidean distance and cosine similarity power clustering & recommenders.

5

Optimisation

Gradient descent uses vector calculus to update model weights.

Vector Notation

Bold lowercase

a, b, v

Arrow notation

$\vec{v}, \vec{a}, \vec{b}$

Component form (2D / 3D)

$(x, y) \quad (x, y, z)$

Column matrix form

$\begin{bmatrix} x \\ y \end{bmatrix}^T \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix}^T$

Unit vector form

$a = ax \, i + ay \, j + az \, k$

Magnitude and Direction

Magnitude (Euclidean norm)

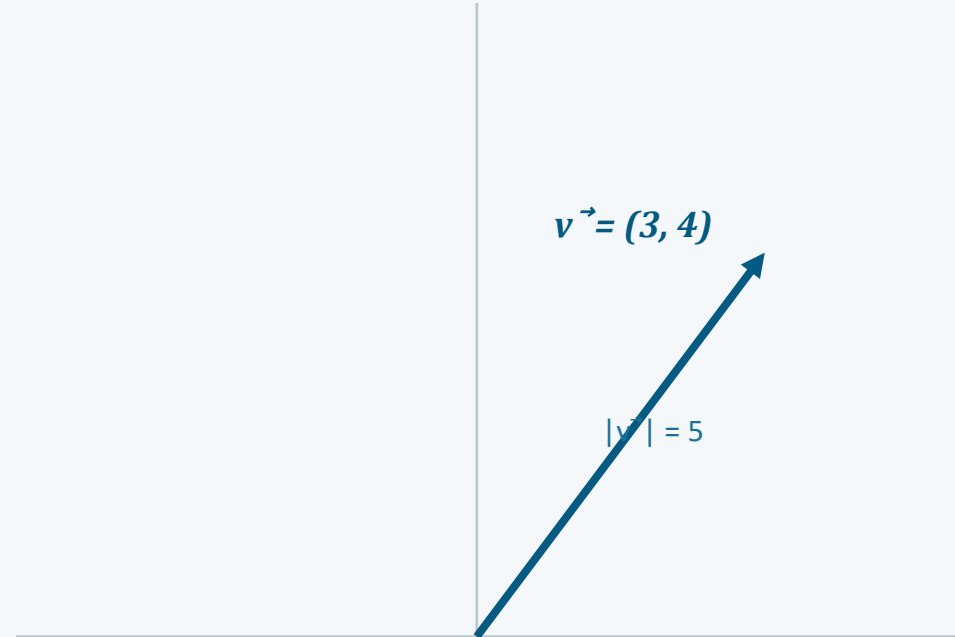
$$|\vec{v}| = \sqrt{x^2 + y^2 + z^2}$$

Direction (2D angle with x-axis)

$$\theta = \arctan(v_y / v_x)$$

Note: consider the quadrant of (v_x, v_y) when reading ϑ .

Worked example: $\vec{v} = (3, 4) \rightarrow |\vec{v}| = \sqrt{9+16} = 5$



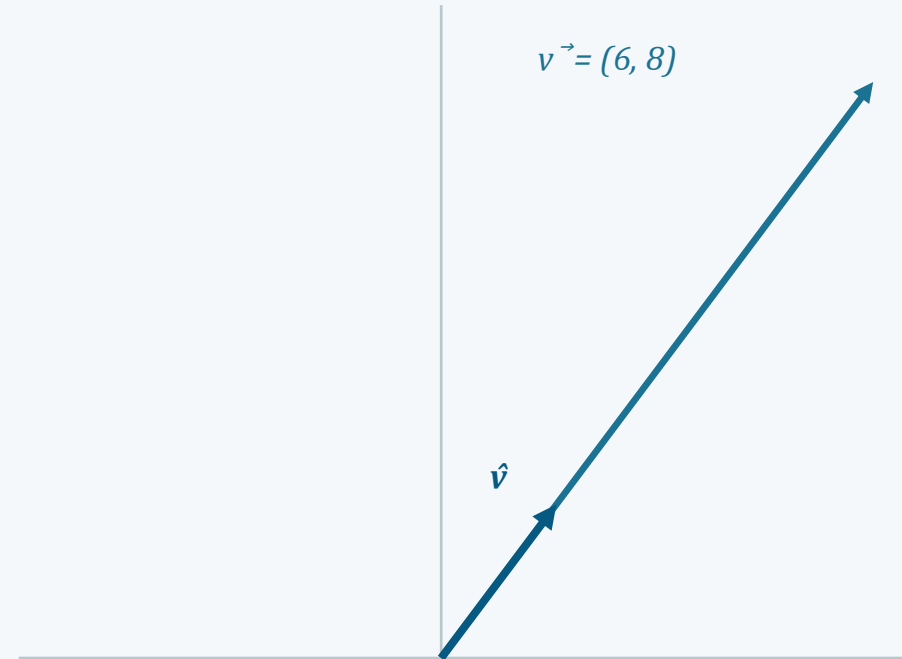
The Unit Vector

A unit vector has magnitude 1 and points purely in the direction of the original vector.

$$\hat{\mathbf{v}} = \mathbf{v}^{\rightarrow} / |\mathbf{v}^{\rightarrow}|$$

Example: $\mathbf{v}^{\rightarrow} = (6, 8)$

$$|\mathbf{v}^{\rightarrow}| = \sqrt{(36+64)} = 10 \rightarrow \hat{\mathbf{v}} = (0.6, 0.8)$$



Vector Addition & Subtraction

Addition — component-wise

$$\mathbf{a}^{\rightarrow} + \mathbf{b}^{\rightarrow} = (\mathbf{a}_x + \mathbf{b}_x, \mathbf{a}_y + \mathbf{b}_y)$$

$$\begin{aligned}\mathbf{a}^{\rightarrow} &= (3, 4), \mathbf{b}^{\rightarrow} = (1, 2) \\ \mathbf{a}^{\rightarrow} + \mathbf{b}^{\rightarrow} &= (3+1, 4+2) = (4, 6)\end{aligned}$$

Geometrically: place $\mathbf{b}^{\rightarrow}$'s tail at $\mathbf{a}^{\rightarrow}$'s head (tip-to-tail rule).

Subtraction — component-wise

$$\mathbf{a}^{\rightarrow} - \mathbf{b}^{\rightarrow} = (\mathbf{a}_x - \mathbf{b}_x, \mathbf{a}_y - \mathbf{b}_y)$$

$$\begin{aligned}\mathbf{a}^{\rightarrow} &= (5, 7), \mathbf{b}^{\rightarrow} = (2, 3) \\ \mathbf{a}^{\rightarrow} - \mathbf{b}^{\rightarrow} &= (5-2, 7-3) = (3, 4)\end{aligned}$$

Geometrically: $\mathbf{a}^{\rightarrow} - \mathbf{b}^{\rightarrow}$ points from $\mathbf{b}^{\rightarrow}$'s head to $\mathbf{a}^{\rightarrow}$'s head.

Scalar Multiplication

$$k \vec{v} = k(v_x, v_y) = (k v_x, k v_y)$$

$$k > 1$$

Magnitude increases, direction unchanged

$$0 < k < 1$$

Magnitude decreases, direction unchanged

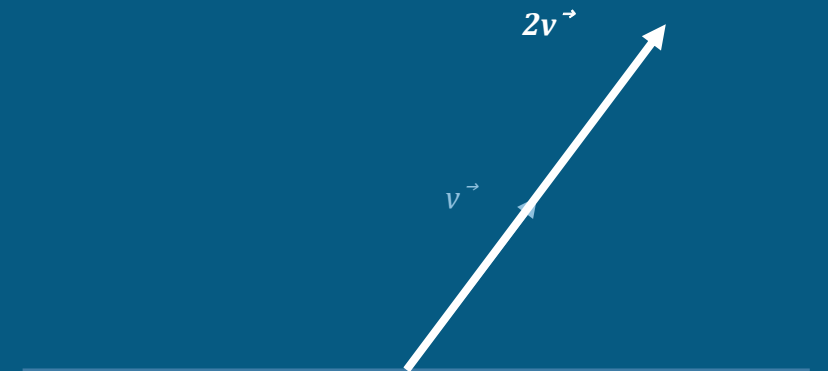
$$k < 0$$

Direction reverses, magnitude scales by $|k|$

Example

$$\vec{v} = (3, 4), k = 2$$

$$2\vec{v} = 2(3, 4) = (6, 8)$$



Dot Product — Definition

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y$$

Alternative form:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \vartheta$$

Worked Example

$$\mathbf{a} = (3, 4), \mathbf{b} = (1, 2)$$

$$\mathbf{a} \cdot \mathbf{b} = (3)(1) + (4)(2) = 3 + 8 = 11$$

Reading the sign of θ

$$\mathbf{a} \cdot \mathbf{b} = 0 \rightarrow \mathbf{a} \perp \mathbf{b} \text{ (perpendicular)}$$

$$\mathbf{a} \cdot \mathbf{b} > 0 \rightarrow \theta < 90^\circ \text{ (acute)}$$

$$\mathbf{a} \cdot \mathbf{b} < 0 \rightarrow \theta > 90^\circ \text{ (obtuse)}$$

Dot Product — Properties & Applications

Properties

- Commutative: $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
- Distributive: $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$
- Orthogonality test via $\vec{a} \cdot \vec{b} = 0$

Applications

- Finding the angle between vectors
- Projection of one vector onto another
- Similarity measure between data vectors
- Neural networks: input · weight at every neuron

Cross Product — Definition

For $\vec{a}=(a_x,a_y,a_z)$, $\vec{b}=(b_x,b_y,b_z)$:

$$\vec{a} \times \vec{b} = (a_y \cdot b_z - a_z \cdot b_y, a_z \cdot b_x - a_x \cdot b_z, a_x \cdot b_y - a_y \cdot b_x)$$

Computed via the determinant of the i, j, k matrix formed by $\vec{a}$ and $\vec{b}$.

Worked Example

$$\vec{a} = (1, 2, 3), \vec{b} = (4, 5, 6)$$

$$\vec{a} \times \vec{b} = (2 \cdot 6 - 3 \cdot 5, 3 \cdot 4 - 1 \cdot 6, 1 \cdot 5 - 2 \cdot 4)$$

$$= (-3, 6, -3)$$

The result is a new vector, perpendicular to both $\vec{a}$ and $\vec{b}$.

Cross Product — Properties & Applications

Properties

- Anti-commutative: $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$
- Result is perpendicular to both $\vec{a}$ and $\vec{b}$
- $|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$
- Parallel vectors $\rightarrow \vec{a} \times \vec{b} = \vec{0}$

Applications

- Finding normal vectors to planes
- Area of parallelograms & triangles
- Reinforcement learning: physics of motion & collisions in simulated environments
- Computer vision: 3D modelling & pose/orientation estimation

Linear Combination of Vectors

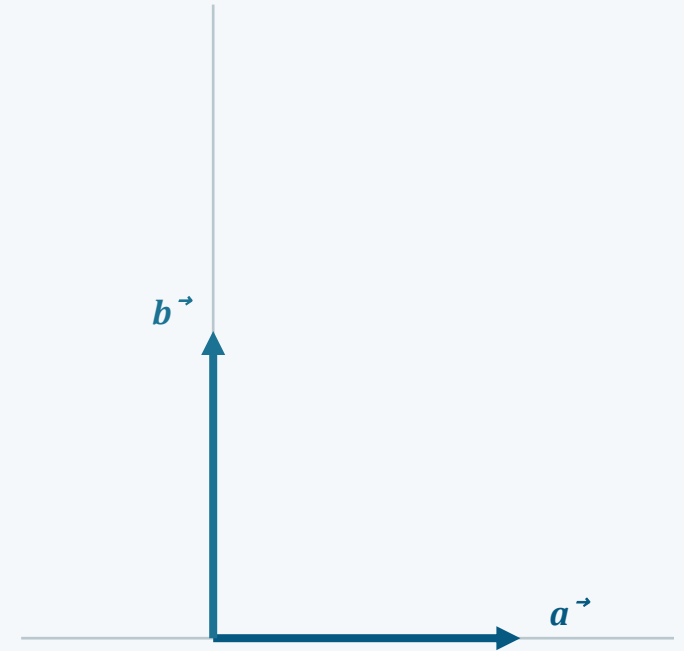
A vector $\vec{v}$ is a linear combination of $\vec{a}$ and $\vec{b}$ if:

$$\vec{v} = c_1 \vec{a} + c_2 \vec{b}$$

Example: standard basis $\vec{a}=(1,0)$, $\vec{b}=(0,1)$

Any vector $\vec{v}=(x,y)$ can be written as $\vec{v} = x \cdot \vec{a} + y \cdot \vec{b}$,
since $c_1 = x$ and $c_2 = y$ solve the system exactly.

$$\vec{v} = x \vec{a} + y \vec{b}$$



Applications of Vectors

1

Physics

Representing force, velocity
and acceleration.

2

Engineering

Direction and magnitude of
physical quantities.

3

Computer Graphics

Position and motion of objects
on screen.

4

Navigation

Determining direction and
displacement.

WORKED EXAMPLE 1

Force Equilibrium

Problem

Three forces act on an object: $\vec{F}_1 = (5, 3)$ N, $\vec{F}_2 = (-2, 4)$ N, $\vec{F}_3 = (a, b)$ N.

Find $\vec{F}_3$ for equilibrium.

For equilibrium: $\Sigma \vec{F} = \vec{0}$

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = (0, 0)$$

$$(3+a, 7+b) = (0, 0)$$

$$\rightarrow \mathbf{a = -3, b = -7}$$

Solution

$$\vec{F}_3 = (-3, -7) \text{ N}$$

This third force exactly balances $\vec{F}_1$ and $\vec{F}_2$, bringing the net force to zero.

WORKED EXAMPLE 2

Vector Projection

Problem

Find the projection of $\vec{a}=(4,3)$ onto $\vec{b}=(1,2)$.

$$\text{proj}_{\vec{b}} \vec{a} = (\vec{a} \cdot \vec{b} / |\vec{b}|^2) \vec{b}$$

$$\vec{a} \cdot \vec{b} = (4)(1) + (3)(2) = 10$$

$$|\vec{b}|^2 = 1^2 + 2^2 = 5$$

$$\text{proj} = (10/5)(1,2) = 2(1,2) = (2, 4)$$

Result

$$\text{proj}_{\vec{b}} \vec{a} = (2, 4)$$

This is the “shadow” $\vec{a}$ casts onto the direction of $\vec{b}$ — the component of $\vec{a}$ that lies along $\vec{b}$.

Cosine Similarity for Recommendations

Problem

User A: $\vec{A}=(5,3,4,2)$ User B: $\vec{B}=(4,4,3,3)$
Find their similarity using cosine similarity.

$$\text{similarity} = (\vec{A} \cdot \vec{B}) / (|\vec{A}| |\vec{B}|)$$

$$\vec{A} \cdot \vec{B} = 20 + 12 + 12 + 6 = 50$$

$$|\vec{A}| = \sqrt{54}, \quad |\vec{B}| = \sqrt{50}$$

$$\text{similarity} = 50 / \sqrt{2700} \approx 0.96$$

Interpretation

$$\approx 0.96$$

Cosine similarity ranges from -1 to 1 . A value near 1 means the users' preference vectors point in nearly the same direction — highly similar tastes, which recommender systems exploit to suggest items.

Exercise — Basic Operations

1

Find the magnitude of $\vec{v} = (-3, 4)$.

2

Given $\vec{a}=(3,-2)$, $\vec{b}=(1,4)$: find (a) $\vec{a}+\vec{b}$ (b) $\vec{a}-\vec{b}$ (c) $3\vec{a}-2\vec{b}$ (d) $|\vec{a}|$ and $|\vec{b}|$

3

Find the unit vector in the direction of $\vec{v} = (6, 8)$.

4

Calculate the angle between $\vec{p}=(1,1)$ and $\vec{q}=(1,-1)$.

5

Find a linear combination of $\vec{a}=(2,0)$, $\vec{b}=(0,2)$ to express $\vec{v}=(6,4)$.

6

Write a short note on vector multiplication (dot and cross product) with examples.

Exercise — Applications & Advanced Operations

APPLICATIONS

Particle moves $A(2,3) \rightarrow B(7,8)$. Find the displacement vector and its magnitude.

Forces $\vec{F}_1=(10,0)$ N and $\vec{F}_2=(0,15)$ N act on an object. Find the resultant force and its direction.

Game character at $(100,200)$ moves with velocity $(5,-3)$ px/frame. Position after 10 frames?

ADVANCED (3D)

$$\vec{u}=(2,-1,3), \vec{v}=(1,4,-2), \vec{w}=(-3,0,1)$$

- a) Calculate $\vec{u} \cdot \vec{v}$ — are they perpendicular?
- b) Find $\vec{u} \times \vec{v}$ — are they parallel?
- c) Calculate $(\vec{u} + \vec{v}) \cdot \vec{w}$
- d) Find the magnitude of $\vec{u} \times \vec{w}$

Key Takeaways



Vectors carry magnitude AND direction — unlike scalars.



Magnitude uses the Euclidean norm; direction uses arctan for 2D vectors.



Addition, subtraction and scalar multiplication all work component-wise.



Dot product $\rightarrow$ scalar result; measures alignment/similarity.



Cross product $\rightarrow$ vector result (3D only); gives a perpendicular vector.



Vectors underpin data representation, ML models, and neural network computation.



Questions & Discussion

Chapter 3.2 — Vectors: Concepts and Applications

Instructor: Saptarshi Jana